# How Do Terrorist Cells Self-Assemble: Insights from an Agent-Based Model of Radicalization

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Since the 9/11 attacks, there has been a steady increase in the incidence of self-starter terrorist plots in both Western Europe and the United States. Past studies of self-starter terrorism focused on how persons come to acquire radical beliefs but take the additional step for granted - the transformation of isolated radical individuals into cells or networks that produce violent collective action. To better understand this search and assembly aspect of radicalization we constructed an agent-based computational model of radicalization and then validated its predictions with empirical data. Our work yielded three distinct contributions: 1) we operationalized radicalization with precise network measures and distinguished between four types of radicalization scenarios; 2) we characterized the factors that contribute to radicalization as well as deradicalization; 3) we extended radicalization theory by identifying specific network mechanisms such as occlusion, encapsulation, and magnet-rewiring by which radicals self-organize. Our findings may allow us to distinguish the conditions that lead to small transient cells from conditions that lead to large long-lasting terrorist organizations.

***Keywords:*** *radicalization; political violence; terrorism; self-starter terrorism; agent-based model; social networks*

July 7, 2011

A long-standing question of political violence research has been centered around the process of radicalization - how do individuals and groups develop and act on the belief that violence is a legitimate instrument to achieve political goals (See Crenshaw 1988; Della Porta 1995; Hudson and Majeska 1999; Post, Ruby, and Shaw 2002)? There is considerable work on how radical beliefs are acquired on the individual level (Bandura 1998; Holloway and Norwood 1997; Moghaddam 2005)[1]. However, much less is understood about how individuals with *already developed* radical beliefs find others with similar views and engage in violent collective action. Indeed this process has been largely taken for granted even by scholars who acknowledge the importance of the group for radicalization.

We begin by discussing the issue of home-grown self-starter terrorism and the larger problem of "search and assembly" in radicalization studies. In section two we offer a novel way to both conceptualize and measure radicalization using social network analysis. Section three explains the agent-based model of radicalization used to investigate the process, the choice of parameters, and the principles underlying the model. In section four we present the analysis and findings of the model along with an empirical validation of some of the model's predictions. We also discuss key mechanisms generating the empirical patterns such as occlusion, encapsulation, and magnetic-rewiring. Section five offers a more general discussion of the results and section six suggests future directions in this research program.

**The Challenge of Home-Grown Self-Starter Terrorism**

Scholarship on radicalization experienced a shift in focus from the study of more traditional recruiter-centered radicalization towards the study of self-radicalization. The shift is driven by two recent trends in terrorism: 1) the re-emergence of the so-called "home-grown

terrorism" – terrorist acts that are carried out by groups whose membership is composed entirely or predominantly of the native-born citizens of the country that is being attacked and 2) "self-starter terrorism" – terrorist acts that are carried out by small groups of individuals who don't seem to be recruited, directed, trained, or financed by any existing terrorist organization or state security agency and form more or less spontaneously through the initiative of their members. These trends have been noted by scholars (Kirby 2007; Vidino 2007; Kohlmann 2008; Krueger 2008) as well as security agencies and police organizations ((Silber and Bhatt 2007; Hulst 2006; House of Commons 2006).

Since 9/11, homegrown self-starter terrorism (HST) has become especially pronounced within Islamist terrorism – that is terrorism motivated by Muslim fundamentalism. Al-Qaeda has shifted to a more decentralized form (Sageman 2004; Sageman 2008), and this trend may accelerate following the recent death of its long-time leader. Furthermore, this decentralization has also been noted in affiliate organizations such as Jemaah Islamiyah (Magouirk, Atran, and Sageman 2008). The rise in Islamic home-grown self-starter terrorism is probably driven by the increasing pressure on Al-Qaeda and the poor integration of Muslims in majority non-Muslim states. As a testament to this trend, a large number of plots in the last ten years (in the West alone) were cases of home-grown, self-starter terrorism: Netherlands (Hofstad Group 2003), Britain (London Attacks 2005; Tubes/subway plot 2005, Glasgow Airport plot 2007), Germany (2006 World Cup plot, 2007 US Airbase plot 2007), Sweden (US embassy plot 2006), Canada (2006 truck bomb plot). The United States is no exception. A RAND report identified 46 cases of "self-recruitment" to terrorism between 2001 and 2009, with 13 occurring in 2009 alone (Jenkins 2010, vii). Most of these plots were intercepted but some cells have succeeded in carrying out an attack and inflicting large casualties. The most notable is the July 7$^{th}$, 2005

London Bombings in Great Britain which killed 56 people (including the bombers), and injured more than 700 (House of Commons 2006)[2] and the 2004 Madrid bombings, that killed 191 and injured more than 2000 people[3]. Other "successful" attacks carried out by HST cells outside the West since 9/11 were the 2003 Casablanca bombings in Morocco that killed 45 and injured more than a 100, the 2004 Taba hotel bombings in Egypt that killed 34 and injured 170, as well as attacks in Pakistan and Iraq (see Benjamin and Simon 2005).

Islamist terrorism may be transforming (or expanding) into a model of terrorism long-used by right-wing and environmental terrorist groups, known as *leaderless resistance* (see Kaplan 1997). The most infamous act of right-wing terrorism, the 1995 Oklahoma City Bombing, was an attack carried out by just two people; but they were propelled by a large movement that created, distributed, and legitimated a very specific ideology, through pamphlets, and other literature (and more recently through the Internet). Extreme right wing movements such as the American Nazi Party and Christian Identity came to the conclusion that they did not have the capacity to wage an organized military campaign and that it would be more effective to indirectly encourage sympathizers to initiate "small acts of violent rebellion" (Kaplan 1997). A similar strategy has been adopted by members of the Earth Liberation Front (ELF) which actively encouraged the formation of new and independent cells. ELF's publicist, Craig Rosebraugh, summarized the idea of leaderless resistance in an introductory video: "There's no realistic chance of becoming active in an already existing cell… Take initiative; form your own cell." (Joosse 2007). It is important to note that in both right-wing and radical environmentalist cases there was a large community of sympathizers that never engaged in direct violence but who contributed to the cultural repertoires necessary to sustain the movement. A similar community of "cultural entrepreneurs" sustains the Islamist movement by producing books that provide

highly militant interpretations of Islam, graphic videos of violent victimization of Muslims by non-Muslims[4], and websites that provide a community to interpret and reframe current events. They use these mediated cultural tools to encourage sympathizers who they do not know personally to commit violence on behalf of the movement. This division of labor is deliberate and is critical to sustaining home-grown self-starter terrorism.

Of course not all HST cells have this short-lived and independent nature. Over time some do develop into large and sophisticated terrorist organizations. For example, the Egyptian and Saudi groups that coalesced into Al-Qaeda formed in roughly this way (Wright 2006). Indeed one may distinguish two types of HST perpetrators – cells that remain small, independent, and transient on the one hand and cells that become large, connected, and long-lasting on the other. These two types of radical cell trajectories may be caused by different combinations of social factors and may hence be distinguishable using our model, as we explain later.

**The Search Problem in Radicalization Research**

The assembly of self-starter cells involves two interacting processes: the acquisition of beliefs supporting violence to achieve political concessions and the search process where radicals find one another and reinforce their extreme views. Case studies of self-starter terrorism reveal that the individuals who join groups have some pre-existing beliefs or at least sympathies to the ideology of the group they join[5]. But there is little doubt that additional radicalization happens at the group level as well. However, the search is not performed with equal zeal by all radicals; indeed it is more often the case that some members are more entrepreneurial than others and play a recruiter function, as was the case with Mohammad Sidique Khan and the 7/7 bombers (House of Commons 2006).

A key insight is that radicalization is not necessarily linear in the sense that the process involves a sequence of steps where radicals acquire extreme beliefs independently and then meet like-minded colleagues and then engage in violence. Instead individuals who are merely sympathetic may be influenced to more extreme opinions by their friends after they join the terrorist group (Sageman 2004). Furthermore members in small groups have a tendency to reinforce each other's views and even further radicalize one another by outbidding each other's extreme opinions (See a review of this phenomenon by Isenberg 1986).

Nonetheless we maintain that all home-grown self-starter cells at some point face a search and assembly problem. Understanding the factors that influence how they solve this problem is essential if one is to comprehend the bigger picture of radicalization. Conceptualizing and measuring this bigger picture is the question we turn to next.

## CONCEPTUALIZING AND MEASURING RADICALIZATION

A critical question for our study, and one that has been largely unexplored in previous work, is how to measure radicalization in the context of social networks. The most basic way is to simply count the fraction of radicals within the relevant community. However this approach is problematic because it does not take the relationships between the radicals into account. There may be an increase in the total percentage of radicals but not in the number of connected cells that these radicals form (and vice versa). Because the network configuration plays such a decisive role we developed three measures of radicalization under the framework of social network analysis. Those measures are based on established ideas of social network theory. Combined together they paint a deeper picture of whether radicalization is occurring and how. We then identify distinct radicalization scenarios that are defined based on the different

combinations of the three basic measures.

**Network Measures of Radicalization: Isolation, Clustering and Mean Cell Size**

The first measure, termed *Isolation*, describes the connectivity between radicals and non-radicals. Low isolation represents a condition where radicals are well-connected to the society at large including many non-radicals (Figure 1, boxes A and B). High isolation represents a condition where the radicals constitute an isolated subnetwork with no ties to non-radicals (Figure 1, boxes C and D). High isolation has the greater threat of violence since the radicals experience no restraining influence from non-radicals. A suitable measure of isolation is the difference between the number of radical-to-radical ties ("internal ties") and radical-to-non-radicals ties ("external ties") divided by the total number of ties. This is equivalent to the E-I index (Krackhardt and Stern 1988) multiplied by -1. Note that Isolation varies from -1 (lowest) to 1 (highest) and it increases if the fraction of radicals in the population increases, *ceteris paribus*.

However, the full picture of radicalization requires additional measures because it usually takes more than a single radical-to-radical dyad, i.e. more than two individuals, to form an effective cell. Part of this picture is painted by *Clustering* which measures the degree to which the radical community is organized into triads; in essence whether the community is "closely knit" into cliques that could be the precursors of active terrorist cells. The distinction between low and high clustering can be further seen in Figure 1; boxes A and C represent scenarios where there is very little clustering between the radicals, while boxes B and D represent a scenario where the radicals are highly clustered. To measure *Clustering* we use the $\tilde{C}$ metric from Soffer (2005) applied to the subnetwork containing just the radicals. It varies from 0 (lowest) to 1

(highest) and makes important corrections for the degrees of the nodes.

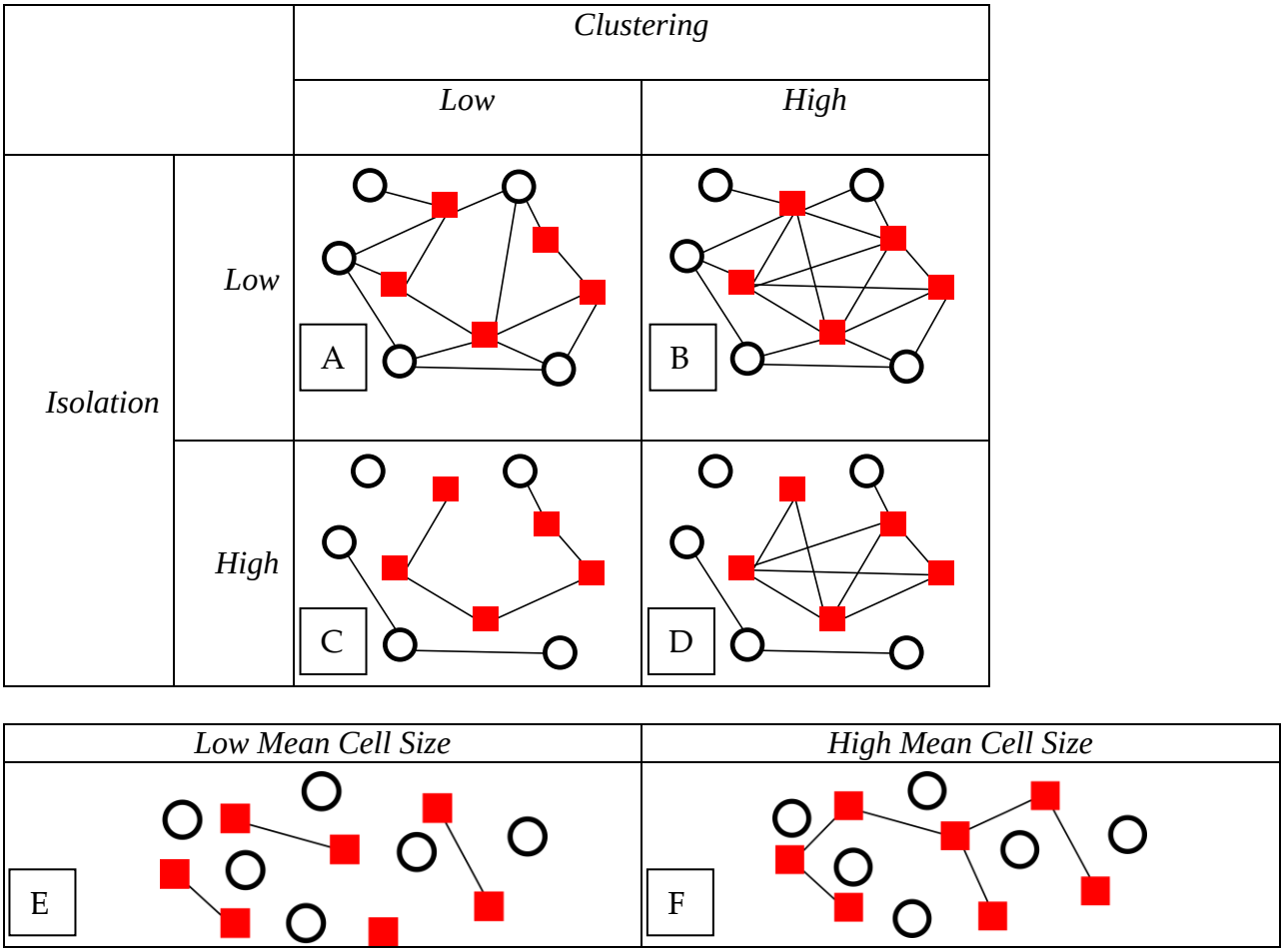



**Figure 1: Three Network-Based Measures of Radicalization: Isolation, Clustering and Mean Cell Size.** When combined the measures define the four radicalization scenarios. The shaded squares represent radicals and the white circles represent non-radicals.

Finally, it is important to consider the scale of the radical community, namely, the sizes of the cells, a metric we call *mean cell size* or just *cell size*. Figure 1 box E illustrates a network with a low mean cell size; while box F illustrates a high mean cell size (while isolation and clustering are the same).

We have also tried alternative measures of clustering and isolation which were highly correlated to the metrics we selected and did not change the analysis. Furthermore, the three metrics we selected show a very low correlation among themselves (see Table 1).

**Table 1**

Correlation Between the Three Metrics of Radicalization in the Simulation Data. Standard errors are shown in parentheses. The table shows that the metrics are largely independent.

| | Isolation | Clustering | Mean Cell Size |
|---|---|---|---|
| **Isolation** | 1 | | |
| **Clustering** | 0.0416 (0.0013) | 1 | |
| **Mean Cell Size** | 0.0264 (0.0408) | -0.0617 (0.0000) | 1 |

These three measures provide information about the state of the artificial society at a particular point in time. In a dynamic society or a social simulation the values of the metrics change over time as ties form and break. In computing the measures in this dynamic context, we therefore averaged them over the course of the simulations[6].

**Types of Radicalization and De-radicalization**

Using the metrics one can characterize two radicalization and two de-radicalization scenarios, summarized in Table 2. Each scenario is populated by certain kinds of radical formations, is associated with a specific kind of terrorism, and can be precisely operationalized in terms of the three radicalization metrics. The four scenarios do not form an exhaustive list and we selected them as ideal types based on empirical cases of HST.

**Table 2**

Definitions of the Radicalization Scenarios. The percentiles refer to the simulation data.

| Scenario | Isolation | Clustering | Mean Cell Size |
|---|---|---|---|
| **Lone Wolfs** | Top 50% | Bottom 50% | Bottom 50% |
| **Wolf Packs** | Top 50% | Top 50% | Top 50% |
| **Trapped Wolfs** | Bottom 50% | Bottom 50% | Bottom 50% |
| **Trapped Wolf Packs** | Bottom 50% | Bottom 50% | Top 50% |

The *Lone Wolfs* scenario is populated by radical individuals organized into small cells that are highly isolated from the rest of society. They are not embedded in groups of like-minded individuals, and have very few associates of any kind. They may correspond to individuals such as Ted Kaczynski, Timothy McVeigh, or Eric Rudolph. Operationally a world of such individuals, in terms of our metrics is characterized by high isolation, low clustering, and a low cell size. In this scenario, the bulk of the terrorism risk comes from Lone Wolfs rather than larger formations, even if many or most of the radicals would never "bite", i.e. pick up arms and commit violence.

The *Wolf Packs* scenario is populated by radicals who form large, internally-clustered cells that are isolated from the rest of society. Wolf packs enjoy the typical benefits of collective action, perhaps producing some cells that are able to execute relatively sophisticated, high-casualty attacks that are unfeasible if each radical operated alone. They correspond to cells such as the 7/7 bombers as well as the Madrid and Hofstad groups. Operationally a world of such individuals, in terms of our metrics, is characterized by high isolation, high clustering, and a large cell size.

The *Trapped Wolfs* scenario is populated by lone radicals who are embedded in a network of non-radicals. They may be individuals who hold radical beliefs and perhaps even contemplated violent action, but who have a dense network of friends indifferent to radical ideology or actively opposing it. These could be relatives or dependents that impose familial obligations on the radicals. For example the typical trapped wolf/radical may wish to act out his beliefs but it would mean making his family destitute and bringing shame on his kin. Once again the claim is probabilistic: a trapped wolf may still "escape" but this escape will be less likely under conditions of being embedded. Operationally a world of such individuals, in terms of our

metrics, is characterized by low isolation, low clustering, and a low cell size.

The *Trapped Wolf Packs* scenario is populated by radical groups who are embedded in a network of non-radicals. This is likely the most common empirical scenario for Islamist groups in Western countries today. As discussed earlier, we know from polling data that there are very large numbers of persons who hold highly extreme beliefs; some of whom coalesce into groups. But what prevents them from acting on those beliefs? A likely explanation is a dense network of ties and obligations to other members of their community who are not radicals: either co-workers, friends, family members, significant others, neighbors, and community leaders. Operationally a world of such individuals, in terms of our metrics, is characterized by low isolation, high clustering, and a large cell size.

# METHODOLOGY

## The Agent-based Modeling Approach

Terrorist cell formation can be investigated using a variety of methodologies, each having a set of strengths and weaknesses. Traditionally, this question has been addressed through historical analysis of defunct terrorist groups, interviews of captured members regarding how they met one another, and detailed studies of autobiographies of terrorists reflecting on the early days of their organizations. These methodologies can provide a wealth of detailed, context-rich information, but they also have a number of limitations. One issue is the simple paucity of reliable data on cell formation, another - symptomatic of qualitative research generally, is the difficulty of separating out general processes from context-specific ones, identifying causal mechanisms, and generalizing beyond the cases examined.

We employ yet another research strategy – one of using empirically-based computer simulations. Simulation methods make it possible to describe interconnected microscopic processes in great detail and then to observe their macroscopic effects. This makes simulations well-suited for studying psychological and social processes, because these are often related in non-linear ways and cause long-run cumulative effects (Macy and Willer 2002). Therefore, simulation methods should be very fruitful in studying how radical groups self-organize because a large portion of this behavior appears to be driven by local interaction of agents, who are also affected in complex ways via ecological variables. Furthermore, by creating artificial societies one can systematically manipulate the parameters of interest to discern meaningful relationships and isolate factors that will be influential over the long-term.

The use of agent-based models to study radicalization is not new: similar works include the Civil Violence Model (Epstein 2002), the Threat Anticipation Model (MacKerrow 2003), NetBreaker (North, Macal, and Vos 2004), Seldon (Berry et al. 2004) among others (Rousseau and van der Veen 2005; Hammond and Axelrod 2006). Our model is different in several respects, most importantly in its use of the Hopfield network (Hopfield 1982; Hopfield 1984; Macy et al. 2003; Nowak and Vallacher 1998) and its focus on the formation of cells primarily around *existing* radicals. The model is also one of the only fully disclosed, open-source code models in the radicalization literature.

**Model Parameters**

How do individuals with radical beliefs find one another and form a group to carry out terrorist attacks, in the absence of a pre-existing organization or a coterie of recruiters? What sort of factors and processes are relevant to model this dynamic? To answer this question, one has to

recognize that the way individuals with radical beliefs find one another and form groups is (at least on the most basic level) no different from the way any set of persons with similar beliefs or interests find each other and form groups, be it reading groups, transient social movements, bible study societies, or swingers clubs. And since group formation has been studied quite extensively, it is reasonable to use findings from this literature as the scaffolding of our model. We then incorporate features unique to our subject matter - minority populations with clandestine attributes (such as beliefs in violence).

Social groups can form around a great variety of interests or activities, but the process of group formation is shaped by three kinds of constraints: structural, individual, and network-related. Each constraint organizes a set of parameters having to do with the different parts of the social ecology: societal factors (population size, social cleavages, tie-organizing sites), individual factors (fixed and variable traits of individuals), and network factors (the size of personal networks, strengths of relationships, and propensity for transitivity). We will discuss the special parameters of the model, the rationale for their inclusion, and what they represent empirically. The *Technical Description* section will relate the parameters to the dynamic processes that underpin the model such as homophily, influence, and transitivity.

***Structural Parameters***

A sizable literature has called attention to the relationship between the size of human aggregates and various sociological processes (Mayhew and Levinger 1976). Therefore a basic structural factor to take into account is population size which we modeled as a single parameter denoting how many agents are interacting. Within a society, the size of social cleavages is considered critical for inter and intra-group relations (Rytina and Morgan 1982). We thus allowed for a distribution of beliefs in political extremism, beliefs which we classified into three

categories: radicals, moderates, or pacifists. The parameters *fraction initially radicals* and *fraction initially pacifist* (and implicitly, the fraction initially moderate) controlled the distribution of beliefs in persons entering the population. Those beliefs were then subject to change via peer influence during the course of the simulation[7].

Next, we incorporated the presence of sites and various social organizations where people meet and form friendships. This is one of the most important factors in the model. Indeed, a House of Commons report on the 7/7 Bombers states that "A common factor for all 3 – Khan, Tanweer and Hussain – was the social life around the mosques, youth clubs, gyms and the Islamic bookshop in Beeston." (House of Commons 2006, 15). The report further states that "camping, canoeing, white-water rafting, paintballing and other outward bound type activities are of particular interest because they appear common factors for the July 7th bombers and other cells disrupted previously and since" (ibid, 17). Moreover, Mohammed Bouyeri, Jason Walters and Samir Azzouz of the Hofstad group in the Netherlands met at a phone shop "that was used by a group of young Muslims to pray, watch gory tapes of terrorist actions, and talk about their desire to participate in jihad" (Vidino 2007, 582). It is not surprising that would-be radicals would meet one another through preexisting groups and organizations. This view is also consistent with the research mobilization literature (Jenkins 1983; McCarthy and Zald 1977). In our terminology, all sites that attract individuals and rewire their social connections by introducing them to new friendships are referred to as *magnets*[8].

Upon examining case studies of Islamist self-starter cells, there appears to be two types of social sites of where the cell members met: 1) *radical magnets* that attract only individuals with extreme views such as mosques led by radical clerics, informal radical seminaries, bookshops and video stores specializing in fundamentalist material; 2) *neutral magnets*, which are locations

that attract individuals without distinction of political views such as athletic clubs, pubs, or social centers. We also included pacifist magnets – a social site that only attracts pacifists. Indeed, there are a number of organizations that promote coexistence[9] but are unlikely to attract radicals or uncommitted moderates. We included them in order to consider the effectiveness of pacifist sites as counter-radicalizing factors.

In modeling magnets we had to specify two additional properties: *fraction exposed to magnets (FEM)* and the *magnetic encounter rate (MER).* The former indicates the fraction of the population that attends any given magnet. The latter is the probability per visit with which an individual would be introduced to another attendee[10]. In practical terms, the exposure parameter may correspond to the unemployment rate: under conditions of high unemployment, magnets of all types are likely to see greater attendance because individuals would tend to have more time to pursue their interests. The encounter parameter characterizes the willingness of people to try new friendships.

Finally, an important factor to consider in a model of tie formation is the number of individuals who are leaving and entering the community, the *attrition rate*. We suspected that outmigration may have a little-noted but important disruptive effect on the formation of radical cells. An additional parameter in our model, *radical attrition increment (RAI),* allowed us to specifically vary the attrition rate of radicals. We believe that their migration rate may be greater than the population as a whole because radical individuals may find themselves maladapted to prosper in their community, and thus may have an increased tendency to migrate away. Another plausible reason to differentiate the attrition rate for radicals from the overall population is that radicals may get arrested if they act on their beliefs.

***Individual Parameters***

Individuals in our model had two kinds of attributes. First, there are fixed characteristics about a person that cannot be changed easily such as race, gender, nationality, personality, and religion. Second, human beings also have attributes that are variable such as interests, hobbies, and more mundane preferences. The overall variability of attributes in the population is determined by a parameter termed *diversity*. A particular kind of variable attribute is political belief in violence, which we call *zeal*. Zeal ranges from pacifist to radical, i.e. from opposing to supporting political violence.

While in practice individuals may have hundreds of these characteristics, only some of them are likely to be socially salient to the formation of ties in a given community. Though, there may be some variability even within the salient parameters. Therefore, the parameters *fixed attribute salience* and *zeal salience* control the weight of the fixed attributes and that of zeal relative to the variable attributes.

Another individual parameter of the model is *pressurability*: the extent to which individuals conform or adopt variable attributes such as opinions from their friends. The higher the pressurability, the more weight individuals put on the average opinion of their friends in the network. For example, under conditions of high pressurability moderates may become radicals if they are surrounded by many radical friends. There are many sources of variation in pressurability; they may range from individual factors such as one's personality to socio-cultrual factors such as collectivism-individualism[11].

***Network Parameters***

Each person has a degree budget, the maximal amount of friends they can have. Marsden (1987) identified the social networks of Americans from the General Social Survey at a maximum of 5, the average of kin members at 1.5, and that of non-kin at 1.4 persons[12]. New ties are formed during the course of the simulation, but once the total number of ties reaches an individual's degree budget, no more are formed or the weakest one is dropped.

Another important network process is triadic closure – the formation of ties between individuals that share friends (see Davis 1970). On the societal level one can think of this as the propensity to form close-knit communities even among strangers. This propensity may vary from society to society and so we controlled this process by the parameter of *transitivity*. Finally, the strength of newly-formed ties may reflect the trust of strangers in a society, and is governed by the parameter *average initial tie strength*.

**Technical Description of Model Dynamics**

In this section we provide additional technical information about the model. We represent a community of *N* individuals each possessing a set of personal attributes, rules of interaction, and ties to other individuals in the community. All agents are assumed to have the same types of attributes. Of those attributes, some are fixed and some are variable. For simplicity, we assume that attributes can be in one of two states, which we represent as +1 or -1. Also, each individual has a political opinion on the issue of political violence ("zeal"): either strongly supportive ("radical") or strongly opposing ("pacifist") or a centrist position ("moderate"). These states are represented as +1, -1, and 0, respectively.

In addition to their attributes, individuals maintain ties to others, where tie strength varies

continuously in the interval 0 to 1: 0 represents a vanishingly weak tie and +1 represents the closest friendship. However, unlike other Hopfield models, we did not allow ties of repulsion, *i.e.* ties of negative strength.

During the simulation, the states of the variable attributes and zeal are affected by peer pressure. We closely follow other simulations that use the Hopfield-network and assume that the agents find the average state of their friends, and move closer to that average. The probability of moving to that state depends on the strength of their ties to their peers as well as their individual pressurability constant.

With each time step in the simulation ("tic"), ties form or break, as well as grow stronger or weaker, as determined by several social processes, most importantly, through homophily (McPherson, Smith-Lovin, and Cook 2001). Homophily is the process where individuals with common attributes are more likely to form ties. The ties also tend to grow stronger when persons have many common attributes. As an example, suppose there are 6 attributes in i and j (including zeal), and the states of i and j are (1,1,-1,1,1,1) and (1,-1,-1,1,1,1), respectively, where the last 1 indicates that they are both radicals. Furthermore, suppose none of the attributes have elevated salience. Since they agree on 5 attributes and disagree on 1, the ultimate tie strength between them would be (5-1)/6 = 0.66. If the absolute strength of a tie falls below 0.01, it is disconnected. In most simulations we also allowed ties to grow stronger over time, to represent increased familiarity and trust between persons. However, we did not find this process had a significant effect on radicalization, such as for example, acting as a break on new tie formation.

Whereas the above dynamics are closely modeled on previous work in Hopfield networks, magnets, degree budgets, transitive tie formation, and attrition represent extensions of the framework[13]. Upon joining the simulated society, an agent is enrolled to attend some of the

magnets. Then, at each tic this individual/ego has a certain probability of visiting the magnets he/she is enrolled in and thus has a certain probability of meeting one random prospective friend/alter, who is also attending that magnet. A friendship is formed only if the ego and alter are compatible (via homophily) in accordance with the "never swap a good friend" rule: the net agreement between the agents must be positive and each of the potential friends must either have (a) a free degree or (b) a friend who is less agreeable[14]. In this latter case, the ego drops this existing friend/alter in favor of the new alter.

Friendships may facilitate the formation of additional ties with other individuals through *transitivity*: each individual E, who has ties to A1 and A2 (who do not know each other), would act like a magnet and facilitate a tie between A1 and A2. Such a transitive process was often observed in radical cells (Sageman 2004). We allowed a tie to form only at the expense of existing ties of Y1 and Y2 in accordance to the "never swap a good friend" rule above.

Our simulation also considers how demographic attrition affects radicalization. For simplicity, we assume that if an individual leaves the simulation, their friends would lose all ties to them. Attrition is modeled by randomly removing a small fraction of the population at each tic, and replacing them with new individuals that on average have the same demographic characteristics as the rest of the population; thereby maintaining the population size constant.

In order to facilitate critical review and further development, we have made the software and simulation data available online[15]. The source code is freely available for download and modification. It is based on the free open-source Repast 3.0 Toolkit (North, Collier, and Vos 2006).

### Statistical Analysis

In order to identify the most sensitive social parameters for each radicalization and de-radicalization scenario, we generated 6000 different simulated societies. Each society was created based on parameter values that were sampled from parameter distributions which we estimated.[16] The samples were then created using the efficient Latin Hypercube Sampling technique (McKay, Beckman, and Conover 1979; McKay 1992; Kleijnen 2008).

To measure the effect of any parameter *q* on radicalization we conducted a sensitivity analysis (Hamby 1994). We compared two distributions: distribution D' for the values of *q* in simulations that achieved a particular scenario (e.g. *Lone Wolfs*), and distribution D for the rest of the simulations. The difference in the means of *q* was standardized, and the distributions compared using the Mann-Whitney test. A large and significant difference implies that *q* contributes to a given scenario (ibid)[17].

## FINDINGS

### Defanging the Wolves: Operant Mechanisms

One of the most interesting findings, in our opinion, has to do not with the effects of the parameters themselves (discussed below) but with the general mechanisms that facilitate those effects. We identified two highly robust mechanisms which are operant in network deradicalization: *encapsulation* and *occlusion*. Encapsulation is the reduction of isolation whereby radicals are encapsulated with non-radicals. Occlusion is the reduction of clustering by blocking transitive tie formation in the first place. Both mechanisms exploit one of the most powerful parameters constraining human networks, which is the degree budget. As previously mentioned human beings have a limited set of friends that they can maintain strong ties with and

if those slots are filled up, certain other friendships may not form.

Encapsulation is facilitated when the degree budget of the radicals is filled up with non-radicals creating a moderating effect. Since radical beliefs are not the only basis on which persons form ties – other traits may also be salient factors upon which friendship is formed. Therefore some individuals holding radical beliefs are (unbeknownst to themselves) "trapped" in a web of non-radical friends. As we shall see certain parameters systematically facilitate such traps.

Occlusion is facilitated when non-radicals[18] fill up the degree budget of a potential radical and prevent triad closure from obtaining. This process is a bit less intuitive to understand and is illustrated in Figure 2. Let's assume that each node has a degree budget of two. Ordinarily, C would form a tie with B via the principle of transitivity, but because C has a strong tie with U, he does not form the tie with B[19].

As we shall see these two mechanisms play a key role in explaining how the model parameters obtain their effects, particularly in the case of deradicalization.

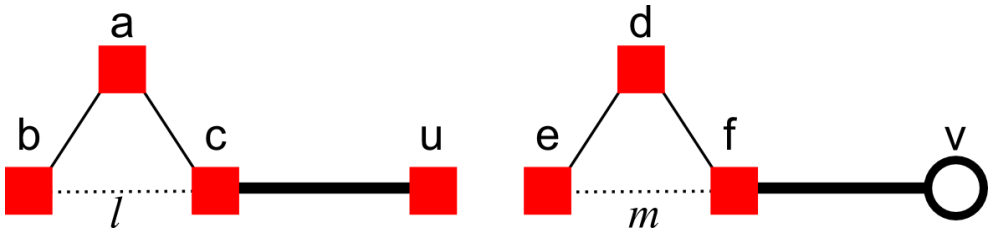



**Figure 2: The Occlusion Process.** The radicals *a,b,c* fail to form a clustered cell (edge l is missing), as may be expected from the triad closure process, because of a strong tie outside the cell between node *c* and node *u.* Node *c* has a tie to a radical (node *u*), which may have formed through an encounter in a radical magnet. A similar process disrupts a triad of *d,e,f* because of *f*'s tie to a non-radical (node v).

**Unleashing the Wolves: Creating Radicalization and Deradicalization**

The data suggests a systematic importance of certain parameters in the production of the four scenarios. While some of the results are consistent with expectation, others are somewhat counterintuitive (in respect to both the positive and negative findings). Greater attention will be paid to the latter type of results. Only statistically significant parameters as well as effect sizes close to or above 0.2 will be considered. The complete summary of the simulation results are shown in Figures 3 and 4. While Figure 3 shows the effects of the parameters on individual metrics of radicalization, Figure 4 shows the effects on each of the four de/radicalization scenarios. For comparison Figure 3 also contains a measure of the fraction of radicals, which is simply a percentage of radicals produced over the course of the simulation and does not take network information into account[20]. Note how the metric is null for most parameters, underscoring the need for metrics that take network information into account.

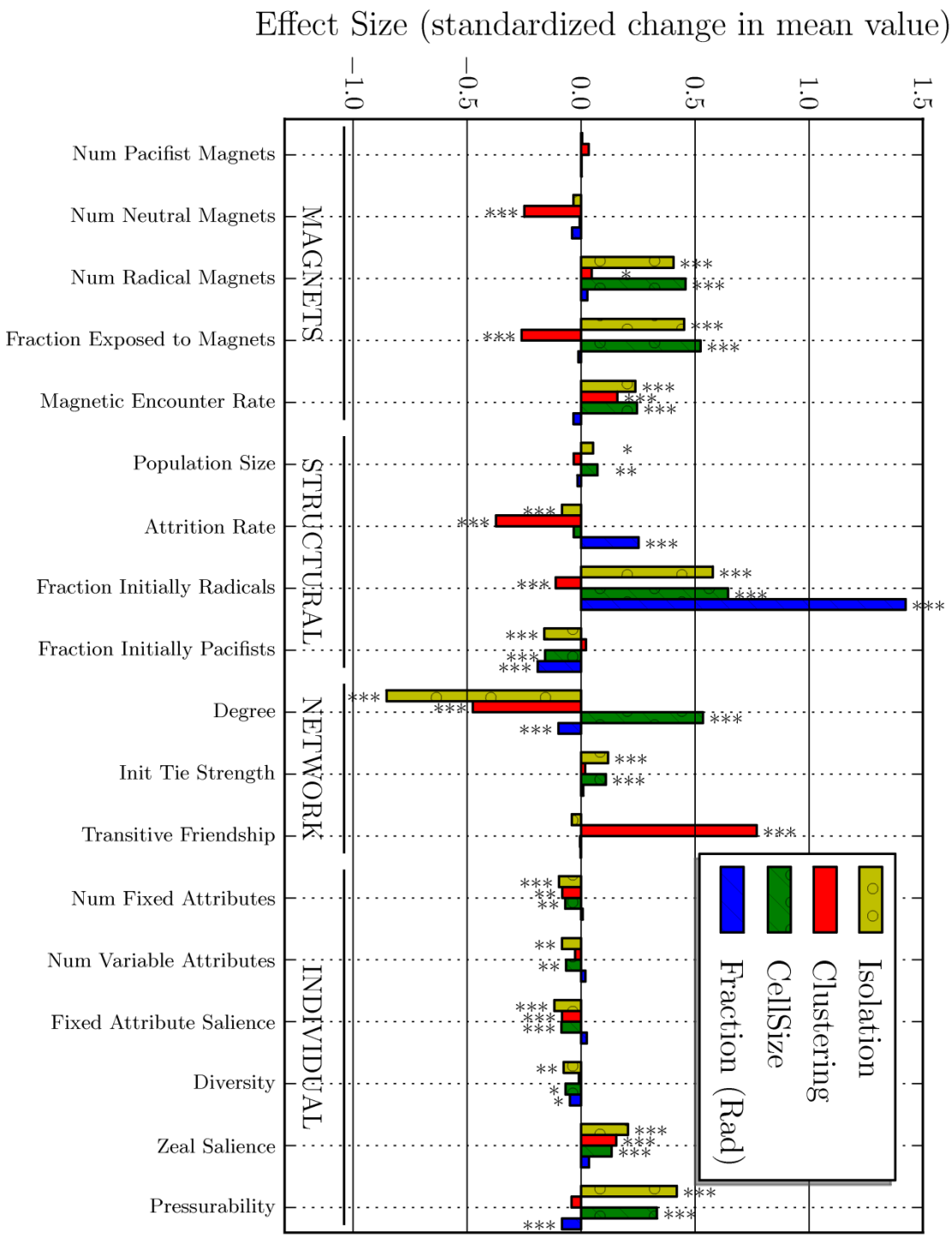



**Figure 3: The Effects of the Model Parameters on the Measures of Radicalization.** The four series represent each of the four measures that were in the top 50% of the simulations. For each parameter, a bar indicates the difference in the mean value of the parameter (mean over the top 50% – the mean in the full set) standardized by the standard deviation in the full set. Significance is based on the Mann-Whitney test (*<0.05, **<0.01, and ***<0.001).

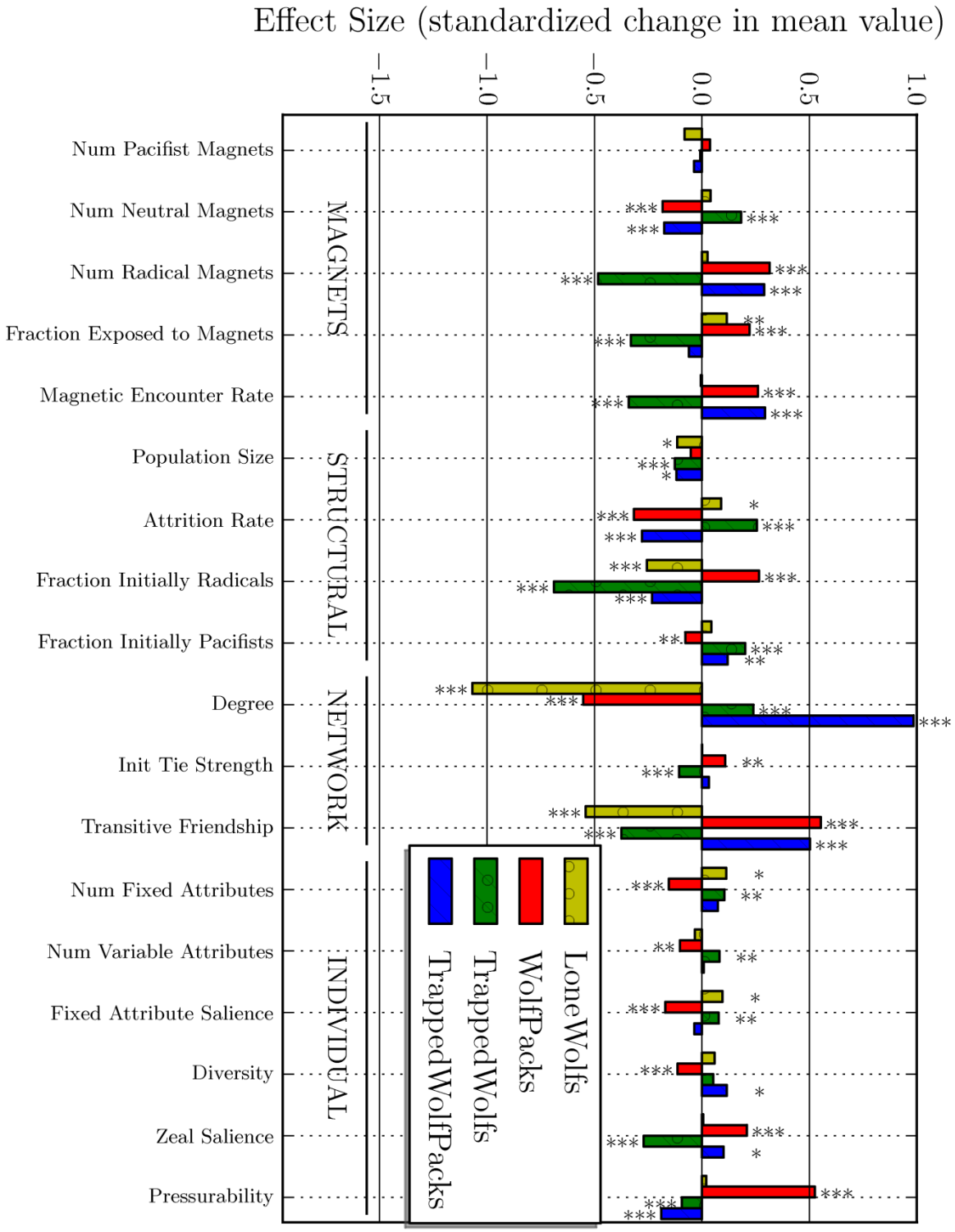



**Figure 4: Parameter-Scenario Sensitivity Analysis.** For each parameter, the bar indicates its importance to the scenario. For each parameter, a bar indicates the difference in the mean value of the parameter standardized by the standard deviation in the full set. Significance is based on the Mann-Whitney test (*<0.05, **<0.01, and ***<0.001).

***Lone Wolfs***

The key parameters associated with creating Lone Wolfs are fraction of initial radicals, average degree, and transitivity. The fraction of initially radicals is negatively related to Lone Wolfs because in a larger radical subpopulation, Lone Wolfs are more likely to coalesce into Wolf Packs. Average degree ("Degree" in Figs. 3 and 4) has a strong negative relationship to Lone Wolfs. The explanation is quite intuitive as having less connections means that one is more likely to be alone. A similar logic applies to transitivity**.** Surprisingly magnets did not contribute to the scenario. One might have imagined that radical magnets would create bonds between isolated radicals (a negative effect on Lone Wolfs). However, magnets are not able to surmount the basic social conditions that lead to Lone Wolfs: a small radical population and a low number of ties (degree).

***Wolf Packs***

The key parameters associated with creating Wolf Packs are radical magnets, fraction exposed to magnets, fraction of initial radicals, transitivity, and pressurability. The role of pressurability is of particular interest. Recall that pressurability controls how easily a node can be "converted" when surrounded by opposite-minded friends. The positive effect of this parameter is counterintuitive because the model did not give any zeal level a stronger appeal/weight than any other. Thus, one might expect that it would be the radicals who are overwhelmed by the much larger non-radical population. Instead the following process likely dominates: radicals form pockets where they are the majority and where they can bring some moderates into their fold.

Increasing the number of neutral magnets, the average degree, or the attrition rate tends to reduce Wolf Packs. Neutral magnets have a negative effect because they introduce radicals to non-radicals, who generally outnumber radicals. The non-radicals occupy the slots on the radical's degree budget thereby reducing isolation (encapsulation) and preventing triadic closure (occlusion). The negative effect of average degree is somewhat surprising, since radical cells themselves require connections to many radicals. While this process does operate, having many friendships also hurts Wolf Packs in two ways. First, the more friends a radical has the more likely one of those friends would be a moderate, reducing isolation; Second, it becomes more difficult to achieve high clustering.

Attrition (population turnover) facilitates the destruction of all social ties, and has the effect of damaging radical packs and reducing radicalization. Radicals who arrive at the community tend to form ties with non-radicals, who are the majority, at least before they travel the long road to finding like-minded persons. The effect of *radical attrition rate* was found to be qualitatively similar to the effect of the overall attrition rate.

### *Trapped Wolfs*

Broadly speaking, Trapped Wolfs are created by the same processes that destroy ties in Wolf Packs or that create ties in Lone Wolfs. Magnets play an important role in the trapping of Lone Wolfs because their number grows with an increase in neutral magnets, a decrease in radical magnets, and a decrease in the fraction exposed to magnets. Structural parameters too play a large role, perhaps affecting this scenario most. Increasing the fraction of initial radicals tends to decrease the number of Trapped Wolfs. This is because the availability of more radicals makes larger cells, ultimately creating Wolf Packs and Trapped Wolf Packs.

Network measures such as average degree exert a positive effect while transitivity exerts a negative effect. Furthermore we see that only one individual factor matters, diversity, and it has a small negative effect. Once again we observe the deradicalizing effect of high average degree and neutral magnets as both parameters facilitate a process of radical encapsulation.

### *Trapped Wolf Packs*

To understand how Trapped Wolf Packs are created, recall that they have high cell size, high clustering, and low isolation. The operant mechanism here is occlusion, which prevents clustering. We see that neutral magnets play an especially prominent role in facilitating occlusion. Wolf packs are trapped not because they are introduced to non-radicals but because radicals are prevented from forming transitive ties to other radicals in the first place.

Much like neutral magnets, attrition exerts a negative role through occlusion rather than encapsulation. It decreases clustering by breaking up radical cells hence converting Trapped Wolf Packs to Trapped Wolfs. Average degree and transitivity on the other hand operate via encapsulation; they facilitate the surrounding of radical cells by non-radicals. Pressurability exerts a negative effect on trapping radical cells because it allows for the easy conversion of moderates who would otherwise act as encapsulators.

## A PARTIAL EMPIRICAL VALIDATION THE MODEL

The robustness of a model depends on the extent to which its predictions correspond to empirical reality. While it is not possible to carry out this type of validation for every model prediction, particularly in respect to our first two measures (e.g. data on the levels of clustering or isolation of radicals are very hard to collect), it is possible to obtain empirical data in respect

to the measure of average cell size. We therefore collected data on cases of HST in the United States and Western Europe from January 1990 through March 2010. Databases such as Lexis-Nexis as well as various lists[21] and datasets[22] were combed for cases of self-starter terrorism. Coders were instructed identify a terrorist cell as self-starter if the following conditions were met: 1) its members are not part (or have not been part of) a professional terrorist organization (PTO); 2) it has not received assistance (e.g. financial, training, or logistical) from a PTO; and 3) it has not received commands/instructions from a PTO. A PTO is defined as a terrorist organization that existed and was active for more than three years, had more than ten members, and engaged in more than three attacks that resulted in casualties or physical damage. The final dataset that we compiled contained 110 cells, ranging from 1 to 13 individuals (237 persons in total) and involved three types of radical movements: Islamic extremism, Environmentalist extremism, and various Right-Wing extremism (e.g. anti-government, hate-crime, anti-abortion)[23]. These movements are the key practitioners of leaderless resistances contrasted to the more traditional organizational/pyramidal terrorism.

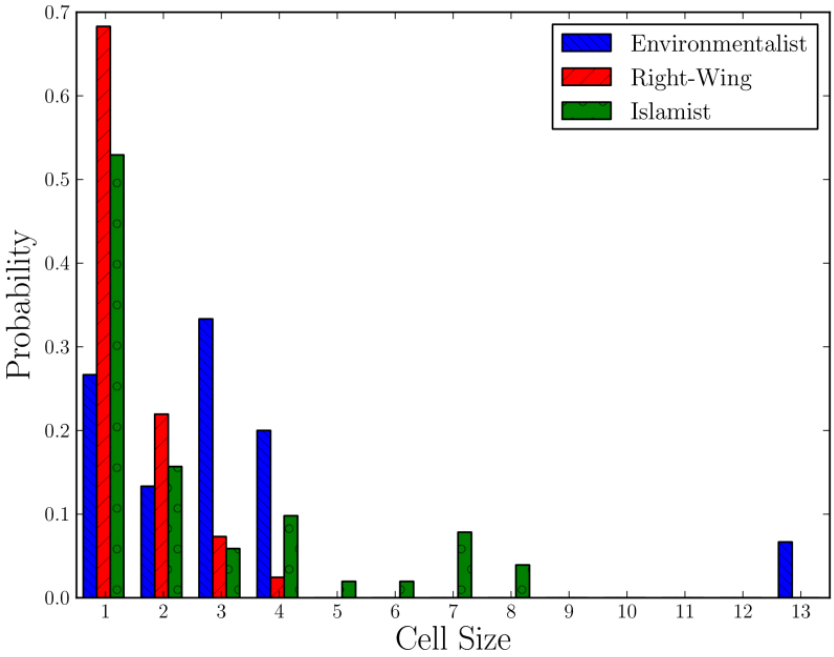



**Figure 5: The Empirical Distribution of Cell Sizes.** Right-Wing cells tend to be small compared to Islamist and especially Envonmentalist cells.

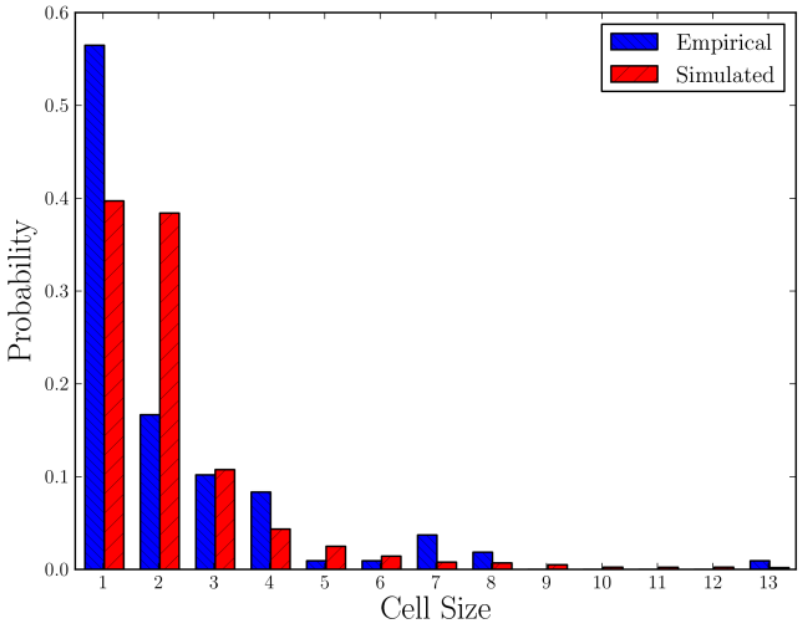



**Figure 6: The Distribution of Cell Sizes in the Simulation and Empirical Data.** The distributions are statistically indistinguishable, giving important validation to the model. Bin 13 contains all cells of size 13 or more.

The average cell size data on those groups is summarized in Figure 5. The figure suggests that Islamist and Right-Wing cells tend to be smaller than the Environmentalist cells (mean cell sizes 2.0, 1.4 vs. 3.2 respectively). The distributions of Right-Wing and Environmentalist cells are significantly different on the Kolmogorov-Smirnov (KS) test ($p<0.01$) but other pairs show no statistically significant difference ($p>0.05$). The KS test was used because it is more robust on small sample sizes. The Mann-Whitney test was in agreement but also identified a significant difference between Right-Wing and Islamist cells ($p<0.05$).

This data was then compared to the predictions generated by the simulations. The goal was to determine whether the average size of the simulated cells approximated the size of the cells observed in the empirical data[24]. Figure 6 shows the empirical and predicted distribution of cell sizes. Visually the simulated data shows fewer 1-person cells and more 2-person cells compared to the empirical values, but otherwise they are quite similar. Indeed, the Kolmogorov-Smirnov two-sample test has a $p=0.49$, suggesting that the distributions are not significantly different (similarly, $p=0.46$ on the Mann-Whitney test). To compute the difference between the distributions, we computed the Hellinger distance score, H (Torgersen 1991). In general, H ranges from 0 to 1; in this case $H = 0.05$, suggesting that the distributions are very close. We also made sure that the result was not a built-in artifact of our estimate for the distribution of average degree: the correlation coefficient of average degree with the mean cell size was only 14%.

The correspondence between the empirical and predicted distributions shows that the model is tapping into reality to some extent. Furthermore it offers suggestive explanations of specific patterns in the sizes of cells between the three types of self-starter groups in our dataset.

Consider Figure 5 - environmentalist groups tend to form larger cells akin to Wolf Packs, while Islamist and rightwing groups tend to form Lone Wolfs. One reason for this trend could be that Islamist groups are often composed of immigrants or their children who lack dense social networks that, by contrast, may characterize activists who makeup environmentalist groups. Right-wing groups who are anti-government may be more anti-society generally and more suspicious of large collectivities. They are less likely to frequent magnets and even less likely than Islamists to have dense social networks; some right-wing terrorists pride themselves on being independent individualists and hence would probably eschew having many friends.

Members of environmentalist movements display the opposite trend, that of being embedded in the broader community and connected to its institutions. This is partly the reason why those groups tend to focus on property damage rather than on injuring or killing, leading to greater societal acceptance of such actions as well as greater support (in our model, greater fraction of initial radicals). Furthermore, because moderate environmentalism is widely accepted in Western societies, extreme environmentalist groups may be able to form and maintain radical magnets with greater ease.

Cell size is also determined by certain tactical requirements. Since environmentalists tend to engage in operations that focus on damaging property, their cells require more people to be effective. In contrast, right-wing Lone Wolfs tend to focus on operations involving firearms and explosives, which do not require such large cells to organize and carry out. A single terrorist wielding a gun can cause many fatalities, while many more people are needed to coordinate and execute a sabotage campaign[25].

## DISCUSSION: TOWARDS AN ECOLOGY OF RADICALIZATION

Our study proposes an ecological perspective in understanding radicalization that considers how the system of social ecology leads to the self-organization of radical networks. This kind of approach complements studies that examine radicalization from a more psychological perspective that has been prominent in the radicalization literature.

The findings from this approach are suggestive for new directions in understanding radicalization and have some important policy implications. An important finding is that radical social sites are very important in contributing to radicalization. At the same time, neutral sites like paintball clubs and sports leagues contribute to *de-radicalization* even though they were thought of as central in explanations of how the terrorist groups formed (e.g. House of Commons Report, 17)[26]. The second finding ran against our prior intuition that any process that promotes homophilous ties would lead to greater radicalization. However, such a supposition ignores the range of possible attributes that can form the basis of ties. Thus, while neutral magnets do form some ties between radicals, their net effect is to de-radicalize because they encapsulate radicals with non-radical ties. Furthermore, pacifist magnets such as "reconciliation societies" do not appear to be effective against radicalization since they draw from people who are already committed to their ideology – that is they essentially preach to the choir.

An important idea that emerges from this study, that may interest policy-makers, is the concept of "disguised magnets". Authorities may want to sponsor neutral magnets but those neutral magnets would be most effective if they are disguised as radical magnets so that they can draw in unsuspecting radicals and gradually integrate them with non-radicals. For example, an organization that offers community paintball – a high action type of activity (that may draw in individuals with radical beliefs), which in subtle ways introduces lessons of coexistence by

pairing Islamists and non-Muslims on the same team. On the other hand radical entrepreneurs may follow the same logic by creating radical magnets disguised as neutral magnets (e.g. the so called "Al Qaeda gym" created by the organizer of the 7/7 cell). Authorities should therefore keep an eye not only on radical magnets but also on neutral ones as well.

Aside from magnets, network density is one of the most critical factors in de/radicalization. If a society wants to prevent radicalization it is not necessary to get the radicals to disavow their beliefs; a more practical objective is to integrate those with radical views by making their networks dense with non-radicals. This strategy requires little effort in trying to find the radicals, and may thus be more cost-effective than an approach based on surveillance and investigative policing.

Pressurability (i.e. impressionability through peer pressure) matters for radicalization. One might have expected that having many pliable individuals would lead to de-radicalization since radicals tend to form a small minority that should be easily overwhelmed by societal norms. Instead, radicals form closely-knit subcultures where they constitute the majority and are able to convert non-radicals that are closely-tied to them.

The results of the model suggest some policies and locations where one may want to focus intervention resources. For example, we found that radicalization is more likely to occur in cities with low migration rates and low number of social venues. Under certain conditions (e.g. low fraction of radicals, high average degree, and low transitivity) Wolf Packs are unlikely to form and the main threat comes from Lone Wolfs. Therefore, when those conditions apply, law enforcement is better focused on individuals that are lone operatives rather than hunting for large groups.

One natural question often raised in the context of our work is the significance of the

Internet for radicalization. Do websites, particularly those that promote radical ideologies, create forums for radicals to exchange information, and make acquaintance with one another; are they a type of "cyber-magnet"? In our opinion websites can serve as magnets only if they facilitate high-trust ties. Arguably, few real friendships are made through such a virtual medium because online persons might be mistrusted and for good reasons: as they can be intelligence officers pretending to be radicals. Radicals would probably want to meet face-to-face to form a tie, certainly if they are to trust one another, and especially if they are hoping to eventually carry out an illegal activity such as an attack. Indeed, we know of virtually no empirical cases where cells were initially formed entirely online. A radical website's main function appears to be to nourish their follower's ideology rather than to help make connections and form cells.

Finally, an important idea that comes out of our work is the notion of radicalization trajectories. Almost every terrorist group was at one point just a "bunch of guys" to use Marc Sageman's widely-repeated phrase (Sageman 2004). What conditions influence whether a cell sprouts into a full-fledged organization or whether it withers away after a few attacks? We refer to this "life-course" of a terrorist cell as its radicalization trajectory. While our model can't make point predictions in this regard, it can shed some light on what the ingredients of the influential conditions look like. A long-lasting organization must initially follow a Wolf Pack scenario: many large, isolated, and clustered cells. We saw that such cells are the result of a high fraction of radicals, low degree budgets, high transitivity, a large number of radical magnets, and high pressurability. Therefore factors that contribute to the increasing of these parameters may nurture the formation of terrorist organizations.

## FUTURE DIRECTIONS

While we have considered a large set of effects, the model is capable of expanding into a number of other directions. In particular it can benefit from adding additional parameters and processes, as well as additional empirical validations some of which we identify below.

Based on our findings it should be possible to construct a risk index for metropolitan areas across Western Europe and the United States. This index will give for each area the likelihood of producing home-grown self-starter terrorist cells, and will be based on the values of the parameters in that area. For this index to be estimated, it would be necessary to obtain a lot of the empirical data, but we believe that the task is feasible because it would be mainly based on existing demographic and economic tables.

The model could also be extended to include other important parameters and processes. One interesting extension is to consider the effect of charismatic individuals (e.g. such as clerics), both radical and non-radical. Unlike magnets who simply introduce individuals, those individuals would increase/decrease the radicalization of individuals who chance upon them. The effect is not predictable in advance because individuals have different susceptibility to such authority figures and may even be repelled by them.

Finally, it is important to point out that in principle the processes we considered are much broader than the phenomenon of self-starter terrorism or even political violence. Indeed in the broad sense they take place whenever any set of individuals who hold extreme views seek one another's company. For example with appropriate adjustments the model and measures of radicalization could be extended to hate groups, cults, marginal social movements, or even gangs and we would like to encourage such applications.

# CONCLUSION

The paper proposed and analyzed a relatively simple, empirically-grounded, agent-based model that treated radicalization as a network ecological process. It showed why, when, and how certain factors such as low attrition, low average network degree, and the presence of certain kinds of meeting sites facilitate de/radicalization, while other plausible factors such as community size have little effect. We hope to have made three key contributions. First, we operationalized radicalization as a network process and distinguished between several kinds of radicalization scenarios. Second, we introduced an agent-based model and characterized the structural, individual, and network factors that contribute to radicalization as well as deradicalization. Furthermore we validated some of the model's predictions empirically. Third; we extended radicalization theory by identifying specific network mechanisms such as occlusion, encapsulation, and magnet-rewiring by which radicals self-organize. We hope to have made contributions to the study of radicalization on both methodological and theoretical grounds; however much more work remains to be done.

## ENDNOTES

[1] We want to emphasize that by "radical" we mean extremist beliefs in supporting the use of violence for political goals. We do *not* mean by "radical" as having any concrete political beliefs (*e.g.* radical vs. reactionary). Hence a radical may be a person who believes in maintaining the status quo by violent means just as much as a person who believes in overthrowing the status quo by violent means.

[2] There is some debate regarding the connection of the Madrid and the London cells to Al-Qaeda. A House of Commons investigation (House of Commons Report 2006:19-20) concluded that there is no evidence for the claim that the July 7th bombing has been directed by Al-Qaeda.

[3] Despite early speculations, a two-year investigation by the Spanish government concluded that there is no evidence for an Al-Qaeda connection(Haven 2006). Additionally not all the collaborators in the Madrid attack were home-grown radicals as some involved were foreign nationals. It is also not clear to what extent a formal organization was behind the attacks.

[4] In a remarkably similar form of dramatizing in-group victimization by the out-group, right-wing extremists vividly describe interracial violence of blacks against whites (Kaplan 1997).

[5] This was clearly the case with the 7/7 Bombers (House of Commons 2006, 15-18); the Hofstad Group (Vidino 2007, 581), and some of the Islamist networks in Spain (Jordan and Horsburgh 2005). Indeed we could find no case of self-starter home-grown terrorism where "brainwashing "or coercion of previously unsympathetic individuals has taken place.

[6] As is common practice with simulation work, the averaging began after a burn-in period of 500 tics to allow the system time to settle into an equilibrium state.

[7] While we sampled a range of parameter space, these parameters can be calibrated based on polling data. For example according to one 2006 UK poll "15% of British Muslims agreed that

violence against civilians, to defend Islam is sometimes justified" (Pew Research Center 2006, 3).

[8] Note that magnets bear some resemblance to foci (Feld 1981). Foci are elements of the social structure that channel association along specific traits resulting in homophily (workplace, neighborhood, city). However, not all foci are magnets. The key property of a magnet is that it has the potential of *re*-wiring ties. Certain foci may "bring homogenous sets of people together" (Feld 1982, 798) but not really change the ties (e.g. neighborhoods, families); instead stabilizing and actually keeping them from being rewired. Second, foci seem to require a pre-existing attribute that organizes people, but such an attribute is not required from magnets. For example, neutral magnets may organize interaction by pure chance (e.g. change meeting at a neighborhood grocery store or a festival). In our conception, a person could visit a neutral magnet based on a kind of focal point strategic reasoning (a la Shelling) or purely out of accidental causes e.g. spontaneous desire for entertainment.

[9] For example the Ansar Youth Project, see http://www.ansaryouth.org.uk/main/

[10] A possible extension of the model would be to allow realistic variation between magnets along those parameters.

[11] Individualistic societies may be characterized as having relatively low pressurability and conformity, compared to collectivist societies (Bond and Smith 1996).

[12] McPherson et al (2006)finds somewhat lower numbers, while studies relying on online social networks or other digital information sources tend to show larger numbers.

[13] Indeed, rather than assuming that all agents are connected to each other (Macy et al. 2003), we explore the connection-forming process in detail. In our simulation, new connections form as a

result of either magnets or introductions by friends.

[14] Old friends/alters receive a bonus based on the longevity of the tie.

[15] The Internet link has been redacted to conform to JCR's policy of anonymizing the manuscript. The software can be transferred electronically to the reviewers (via the editor) if s/he requests it.

[16] Interested readers may consult the attached simulations data to examine the parameter space we used.

[17] In addition to the distribution analysis we examined sensitivity using other techniques. In particular, we fitted rank order regression models and quantile regression models. The most sensitive parameters were identified consistently. The above D-D' technique was chosen since it is most robust to non-linearity and heteroscedasticity.

[18] The process can also theoretically occur with radicals. A radical from a different cluster/cell may occupy the degree budget of another radical preventing triadic closure.

[19] Most people have a degree budget higher than two; this example is meant as the simplest illustration of the process.

[20] This measure may appear highly dependent on the fraction of initial radicals. While this is partly true (the correlation is 45%) other processes contribute to the increase and decrease in the fraction of radicals. In particular attrition and more importantly influence dynamics contribute to the fraction metric in highly non-linear ways.

[21] Southern Poverty Law Center, FBI Chronology of Domestic Terrorism, Anti-Defamation League as well as other sources were used to compile a list of cases.

[22] The datasets we used were compiled by Alan Krueger (Krueger 2008), Steven Brooke and

Robert Leiken (Leiken and Brooke 2006) , American Terrorism Study(TRC 2010), and the Global Terrorism Database (START 2010).

[23] There were a few rare cases of entrapment or plea deals where judgment calls had to be made regarding inclusion. Space limitations in the paper do not allow us to discuss all the criteria used. Interested readers should contact the authors for details.

[24] It is important to recall that "radicals" are defined quite broadly in the simulations - as essentially anyone with a high zeal level, regardless of their involvement in violent activities. Clearly only a fraction of such radicals act on their beliefs and become violent. Hence this comparison is possible only to the extent that the shift to actual violence does not depend on the size of the grouping in a significant way.

[25] We would like to acknowledge that these explanations of difference in cell size of the three self-starter radicalization movements is partly speculative and a more exhaustive study would need to be conducted to conclusively pin down the cause for the size variation.

[26] Also see this popular press article that claims that gyms rather than mosques are the breeding ground for radicalism (O'Neill 2006).

## ACKNOWLEDGEMENTS

The authors are listed alphabetically to indicate equal co-authorship. We would like to thank Michael Macy, David Strang, and Sarah Soule for helpful comments. AG thanks Aric Hagberg and Feng Pan. Parts of this project were funded by the Defense Threat Reduction Agency and the Los Alamos National Laboratory LDRD program. Released as Los Alamos Unclassified Report LA-UR-11-01479.

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